

Limestone is available with the composition 90 wt% calcium carbonate (CaCO_3), 8 wt% magnesium carbonate (MgCO_3), and 2 wt% Insoluble Inerts (I). The limestone is heated to 1000°C in a continuous rotary lime kiln to drive off carbon dioxide (CO_2) and make lime. Lime will include the CaO and MgO produced plus all the impurities present in the limestone which remain after all the CO_2 has been driven off. Assume that both the calcium carbonate and the magnesium carbonate react completely to produce carbon dioxide.

Data: $M_{\text{CaCO}_3} = 100 \text{ kg/kmol}$, $M_{\text{MgCO}_3} = 84.3 \text{ kg/kmol}$, and $M_{\text{CO}_2} = 44 \text{ kg/kmol}$.

- (a) What is the fractional atom economy for the process if lime is the desired product? (Hint: You can choose any convenient basis for your calculation, such as, 100 lb_m of limestone.)
- (b) If limestone costs \$87/ton and lime can be sold for \$125/ton, what is the profit (or loss) rate (\$/day) for a lime kiln producing 250 ton/day of lime? Be sure to indicate whether it is profit or loss. (Hint: How much limestone is required to make 250 ton/day of lime?) 1 ton = 2000 lb_m
- (c) If for this process, it costs \$75/ton of carbon dioxide for the proper disposal of CO_2 waste, how does this change the result in part (b)?

a.) Feed: $\text{CaCO}_3 = 90\text{K}$, $\text{MgCO}_3 = 8\text{K}$, Inerts = 2K $\rightarrow$ 100 lb_m of limestone

Products from CaCO_3 Products from MgCO_3 Lime: $50.4 + 3.82 + 2 = 56.22 \text{ K}$

$90\text{K} \cdot \frac{56}{100} = 50.4$ $8 \cdot \frac{42.3}{84.3} = 3.82$ $\text{CaO} = 40.08 + 16 = 56$

Atom economy = $\frac{56.22}{100} = 56.22\%$ ✓ $\text{MgO} = 24.3 + 16 = 40.3$

b.) $\frac{100}{56.22} = 1.77 \text{ tons of limestone}$ $250 \cdot 1.77 = 442.5 \text{ tons limestone/day}$ ✓

$442.5 \cdot 87 \frac{\$}{\text{ton}} = 38,497.5$ $38,497.5 - 31,250 = 7297.5 \frac{\$}{\text{day}}$ ✓

$250 \cdot 125 = 31,250 \text{ day}$

c.) $75 \frac{\$}{\text{ton}}$